hundred years later, Faraday had found the solution – not for the force of gravity, but for the electric and magnetic forces: the field. Electric and magnetic fields ‘carry around’ the electric and magnetic force.

It’s clear, at this stage, to any reasonable person, that the force of gravity must have its Faraday lines as well. It’s clear also, by analogy, that the force of attraction between the Sun and the Earth, or between the Earth and falling objects, must be attributed to a field – in this case, a gravitational field. The solution discovered by Faraday and Maxwell to the question as to what carries the force must reasonably be applied not only to electricity but also to gravity. There must be a gravitational field, and some equations analogous to Maxwell’s, capable of describing how Faraday’s gravitational lines move. In the first years of the twentieth century this is clear to any sufficiently reasonable person; that is to say, only to Albert Einstein.

Einstein, fascinated since adolescence by the electromagnetic field that pushed the rotors in his father’s power stations, begins to look into this gravitational field and search for what kind of maths could describe it. He immerses himself in the problem. It would take ten years to resolve. Ten years of manic studies, attempts, mistakes, confusion, brilliant ideas, wrong ideas, a long series of articles published with incorrect equations, further mistakes and stress. Finally, in 1915, he commits to print an article containing the complete solution, which he names the General Theory of Relativity: his masterpiece. It is Lev Landau, the most outstanding theoretical physicist of the Soviet Union, who called it ‘the most beautiful of theories’.

The reason for the beauty of the theory is not hard to see. Instead of simply inventing the mathematical form of the gravitational field and seeking to devise the equations for it, Einstein fishes out the other unresolved question in the furthest depths of Newton’s theory and combines the two questions.

Newton had returned to Democritus’s idea, according to which bodies move in *space*. This *space* had to be a large, empty container, a rigid box for the universe; an immense scaffolding in which objects run in straight lines, until a force causes them to curve. But what is this ‘space’ which contains the world made of? What *is* space?

To us, the idea of space seems natural, but it is our familiarity with Newtonian physics that makes it so. If you think about it, empty space is not part of our experience. From Aristotle to Descartes, that is to say, for two millennia, the Democritean idea of space as a peculiar entity, distinct from things, had never been seen as reasonable. For Aristotle, as for Descartes, things have extension: extension is a property of things; extension does not exist without something being extended. I can take away the water from a glass, but air will fill it. Have you ever seen a really empty glass?

If between two things there is *nothing*, Aristotle reasoned, then there is nothing. How can there be at the same time something (space) and nothing?